KEITHLEY 2450

I-V Studio

User Manual

Professional I-V Characterisation · FET Analysis · Q-Point Computation

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1. Introduction

The **Keithley 2450 I-V Studio** is a professional desktop application for instrument-controlled electrical characterisation using the Keithley 2450 SMU. It provides a complete workflow from instrument connection through measurement execution to data export, with a dedicated FET Characterisation module for MOSFET analysis.

This manual covers every panel, parameter, and workflow. It is intended for engineers and researchers who need reliable, repeatable I-V measurements.

1.1 Key Capabilities

- Voltage and current sweeps — linear, logarithmic, and list-based
- Dual sweep (forward + reverse) for hysteresis analysis
- Pulse mode with configurable on/off timing
- Stepper mode — Node 2 steps a bias while the master SMU sweeps
- Real-time live plot with engineering-notation axes
- FET Transfer Curve (Id vs Vgs) and Output Curve (Id vs Vds) via TSP-Link dual-SMU
- **Q-Point analysis:** gm computation (Transfer), load-line + Pd hyperbola (Output)
- Auto-save to CSV + PNG on completion

1.2 System Requirements

Component	Requirement
Operating System	Windows 10 / 11 (64-bit)
Python	3.9 or later
VISA Runtime	NI-VISA 21+ or Keysight IO Libraries Suite
Instrument	Keithley 2450 SMU (firmware 1.7.0+)
Second SMU (FET mode)	Second Keithley 2450, TSP-Link cable
Display	1280 × 720 minimum, 1920 × 1080 recommended
Python Packages	pyvisa, numpy, matplotlib

1.3 Installation

1. Install the VISA runtime from ni.com or keysight.com
2. Install Python dependencies:
```
pip install pyvisa numpy matplotlib
```
3. Run the application:
```
python keithley_iv_studio.py
```
4. To build a standalone .exe:
```
pyinstaller --onefile --windowed --name IV_Studio keithley_iv_studio.py
```

2. Application Overview

2.1 Main Window Layout

On launch the application opens a maximised window divided into: the top header bar, the left scrollable control panel, the right chart area, and a status bar at the bottom.

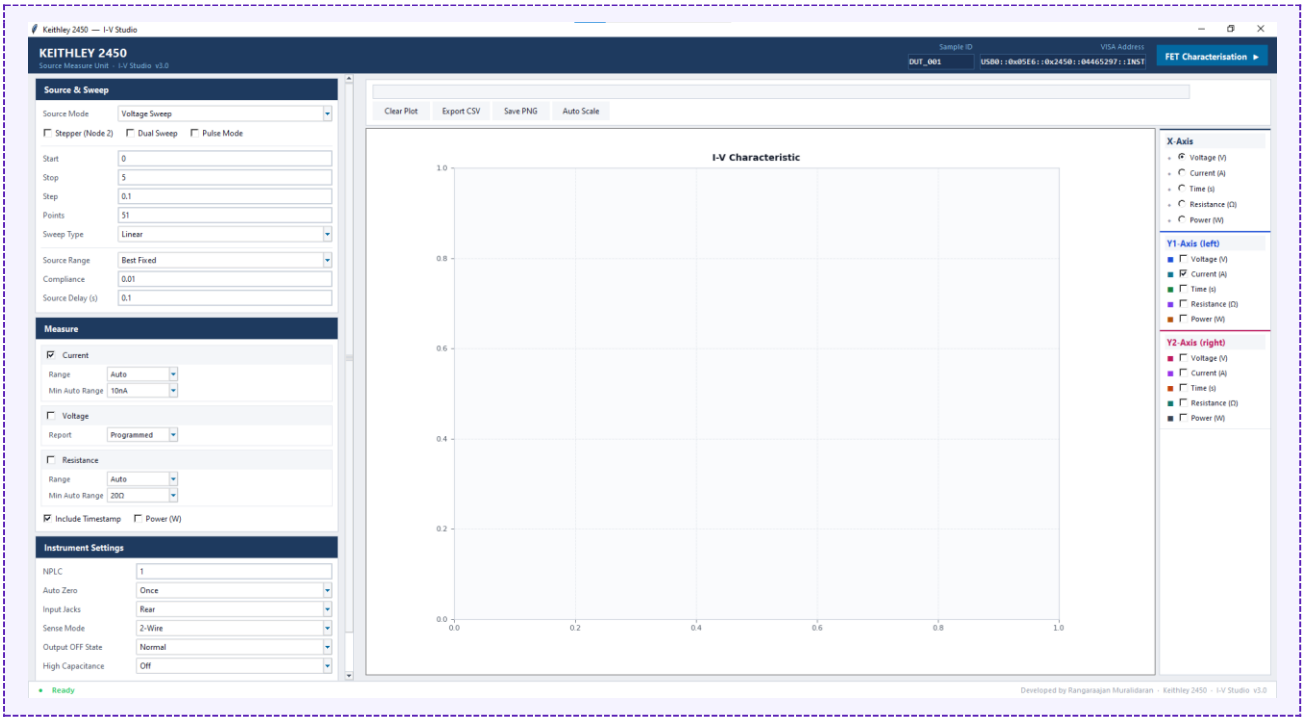

Figure SS-01: Full Main Window

2.2 Header Bar

Element	Description
KEITHLEY 2450 title	Application identifier and version number
VISA Address field	Full VISA resource string for the master SMU — editable at any time
Sample ID field	Short identifier for the device under test — used in auto-saved filenames and CSV headers
FET Characterisation button	Opens the dedicated FET window (see Section 5)

ℹ **Note**	The VISA address must match exactly what NI-MAX or Keysight Connection Expert shows, e.g. USB0::0x05E6::0x2450::04465297::INSTR

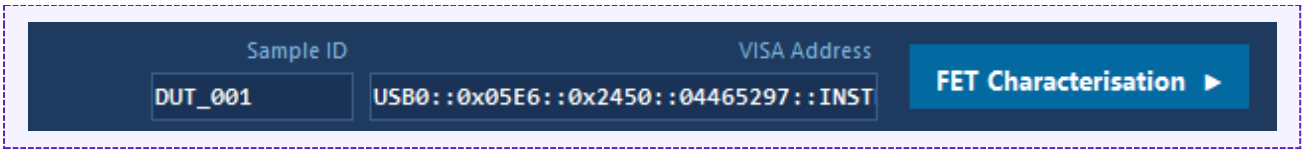

Figure SS-02: Header Bar Close-up

2.3 Status Bar

Indicator	Colour	Meaning
Ready	Green	Instrument idle, ready to sweep
Running	Amber	Sweep in progress
Stopping	Red	Stop requested, waiting for abort
Complete	Green	Sweep finished successfully, data auto-saved
Error	Red	Instrument or communication error
Stopped	Red	Sweep halted by user

3. Left Control Panel

The left panel is scrollable and contains all measurement parameters, grouped into cards. Changes take effect on the next RUN.

3.1 Source and Sweep Card

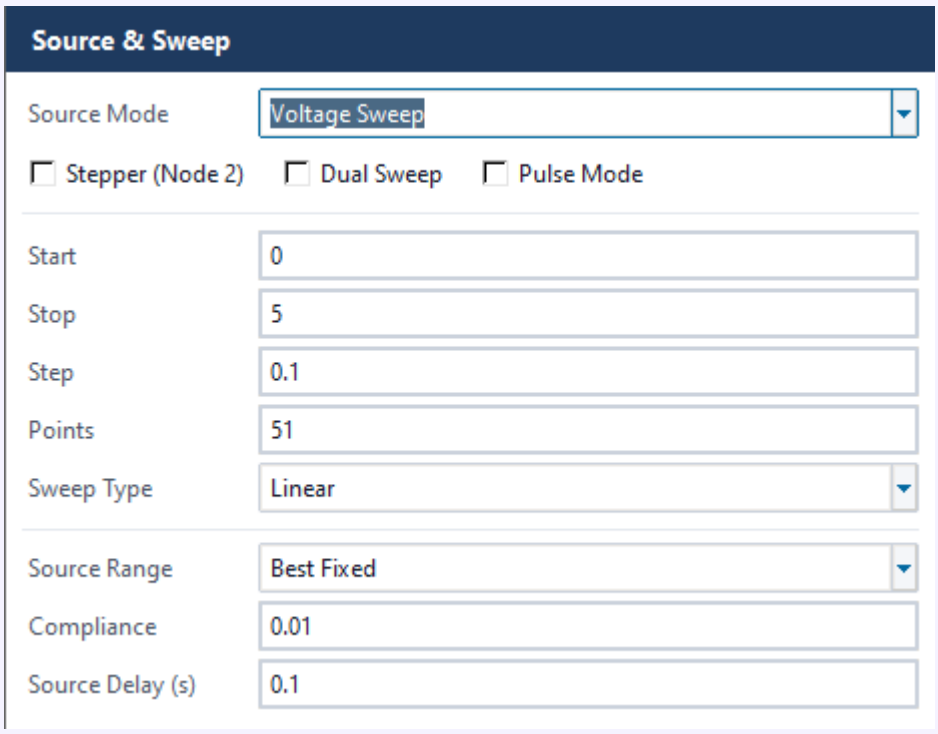

Figure SS-03: Source & Sweep Card — Voltage Sweep Mode

3.1.1 Source Mode

Mode	SMU Output	Typical Use
Voltage Sweep	Voltage (step)	Diode, resistor, semiconductor IV
Current Sweep	Current (step)	LED characterisation
Voltage List Sweep	Voltage (arbitrary)	Non-uniform voltage steps
Current List Sweep	Current (arbitrary)	Non-uniform current steps
Voltage Bias	Single fixed voltage	Leakage, standby current
Current Bias	Single fixed current	VF at fixed current

3.1.2 Option Checkboxes

- **Stepper (Node 2):** Node 2 steps through bias voltages; full sweep runs at each bias — produces a family of curves.
- **Dual Sweep:** Runs sweep in both directions (e.g., 0→5 V then 5→0 V) to reveal hysteresis.
- **Pulse Mode:** Replaces DC sweep with a pulsed sweep — requires On Time and Off Time parameters.

3.1.3 Sweep Parameters (Start / Stop / Step / Points)

Start, Stop, Step, and Points are linked. Editing any three auto-updates the fourth: Points = |Stop − Start| / |Step| + 1.

✓ **Tip**	Sweep Type: Linear gives uniform spacing. Logarithmic gives log-spaced points between Start and Stop — useful for impedance or leakage sweeps spanning decades.

3.1.4 List Sweep Values

When a List Sweep mode is selected, a list editor table replaces the Start/Stop/Step area. Double-click any row to edit inline; use + Row / Delete Row buttons or the Import button to load values from a CSV or Excel file.

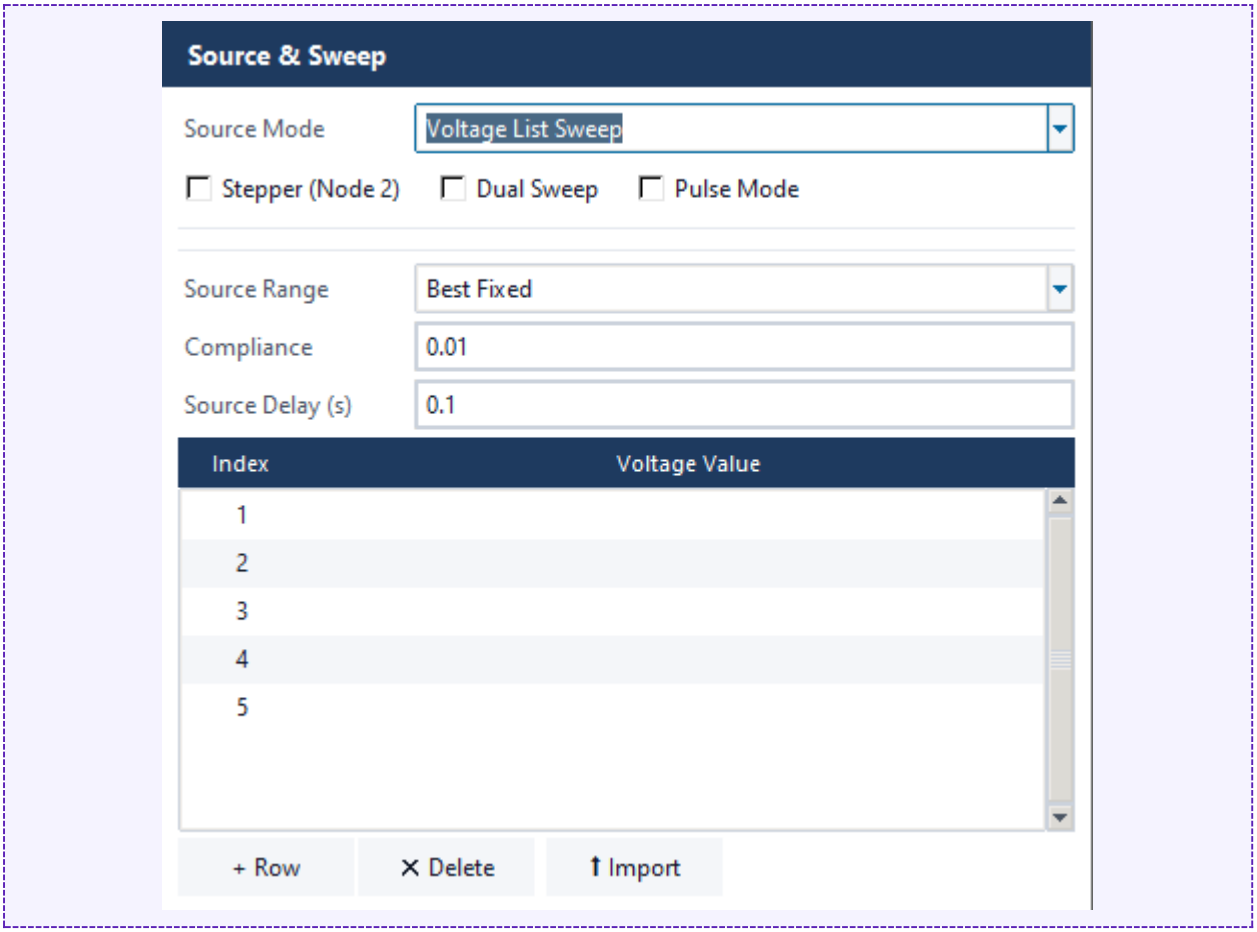

Figure SS-04: List Value Editor Popup

3.1.5 Pulse Parameters

Visible only when Pulse Mode is checked.

Parameter	Description
Bias Level	Voltage applied between pulses — the off-state level
On Time (s)	Duration of each pulse — time at the sweep level per point
Off Time (s)	Duration between pulses — recovery time

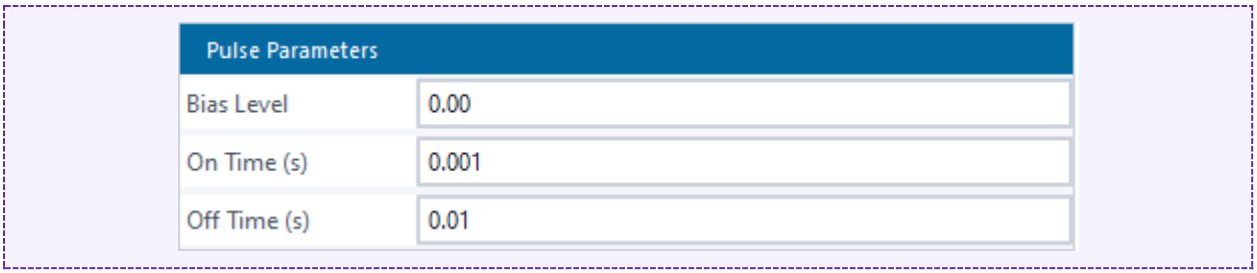

Figure SS-05: Pulse Mode Expanded

3.1.6 Source Range, Compliance, and Source Delay

Setting	Behaviour
Best Fixed	Smallest fixed range covering the full sweep — fastest settling (recommended)
Auto	Auto-ranges point by point — slowest but handles wide dynamic range
Fixed range	Forces a specific range — use when Best Fixed selects an oversized range

⚠ **Warning**	If measured current is flat at an unexpected level the device has hit compliance. Raise the Compliance value or reduce the sweep range.

3.1.7 Stepper — Node 2 Configuration

Visible only when Stepper is checked. Requires a second 2450 on TSP-Link (Master = Node 1, Slave = Node 2).

Parameter	Description
Start (V)	First bias voltage applied by the Slave SMU
Stop (V)	Last bias voltage applied by the Slave SMU
Points	Number of equidistant bias steps between Start and Stop

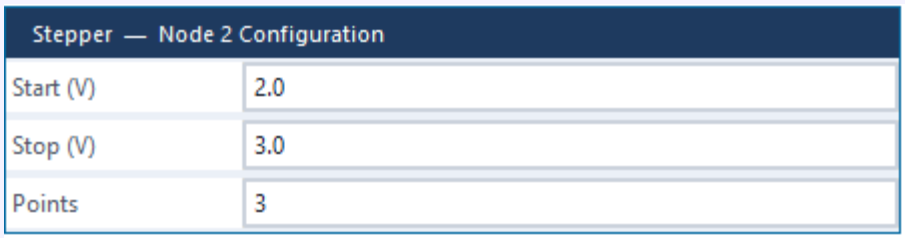

Figure SS-06: Stepper Panel Expanded

3.2 Measure Card

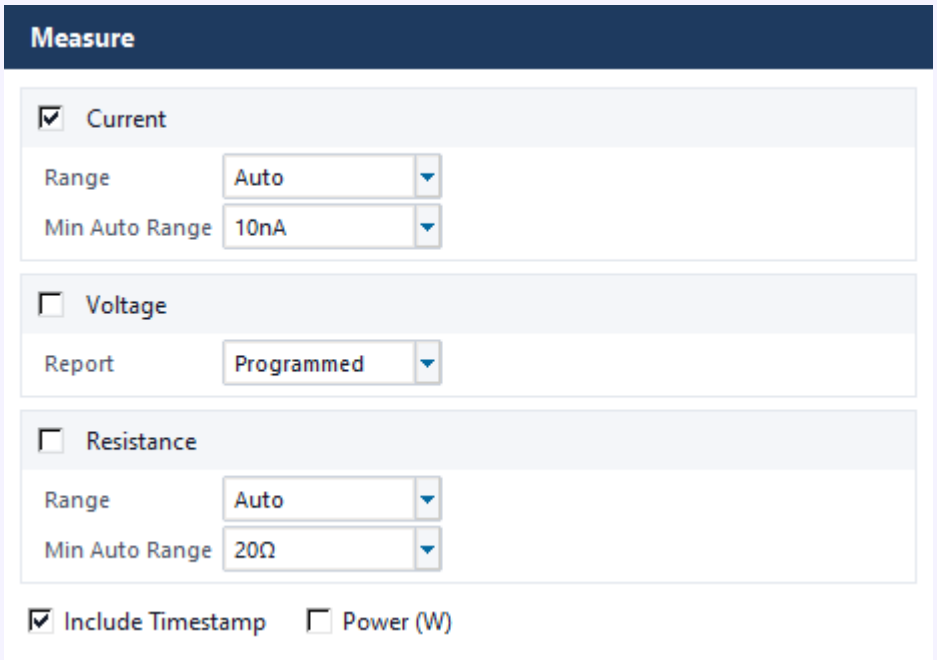

Figure SS-07: Measure Card

3.2.1 Current, Voltage, and Resistance

- **Current Measure:** Check to record drain/output current. Range = Auto uses auto-ranging; a fixed range gives faster measurements at that scale.
- **Min Auto Range:** Prevents the instrument from ranging below this value — avoids slow transitions on near-zero readings.
- **Voltage Measure:** Programmed = commanded voltage; Actual = ADC-measured terminal voltage.
- **Resistance:** Computed as |V/I| per point. Enable for devices where the R-V curve is the primary quantity of interest.
- **Power (W):** Computed as |V×I| per point.
- **Include Timestamp:** Adds elapsed time (seconds since sweep start) as the first CSV column.

3.3 Instrument Settings Card

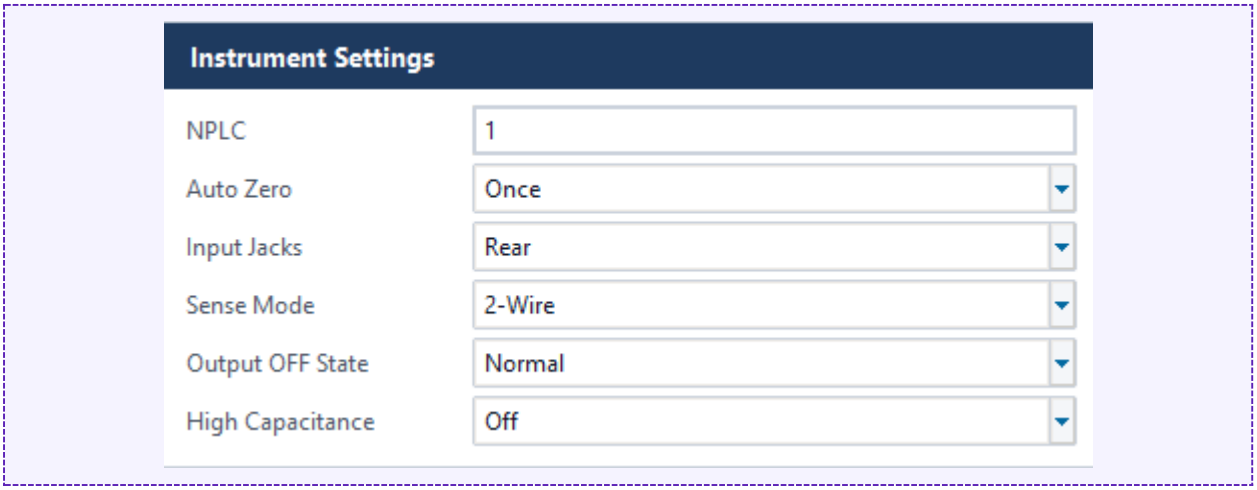

Figure SS-08: Instrument Settings Card

Parameter	Options	Description
NPLC	0.01 – 25	Power Line Cycles per measurement. Higher = more averaging, slower, less noise. 1 NPLC = 20 ms at 50 Hz.
Auto Zero	Once / Off / On	Once: zeros ADC before the sweep (recommended). Off: fastest. On: zeros before every point.
Input Jacks	Rear / Front	Which terminals are used — must match physical wiring.
Sense Mode	2-Wire / 4-Wire	4-Wire (remote sense) eliminates lead resistance — use for low-resistance devices.
Output OFF State	Normal / High-Z / Zero	Behaviour when sweep ends or is stopped. High-Z disconnects output completely.
High Capacitance	Off / On	Enable for devices with capacitance > 10 nF to avoid false tripping.

4. Chart Area and Axis Controls

4.1 Toolbar and Progress Bar

Button	Action
Clear Plot	Clears all sweep data and resets the chart — prompts for confirmation
Export CSV	Opens a save dialog to export the current dataset
Save PNG	Opens a save dialog to save the chart as a 300 DPI PNG
Auto Scale	Resets both axes to fit all visible data

The progress bar fills from 0–100% as data is acquired. The percentage and elapsed time (MM:SS) update in real time.

4.2 Live Chart

New data points appear on the chart as they are collected — the chart refreshes every ~5% of sweep progress. Axes use SI engineering notation (e.g., 10 mA instead of 0.01 A).

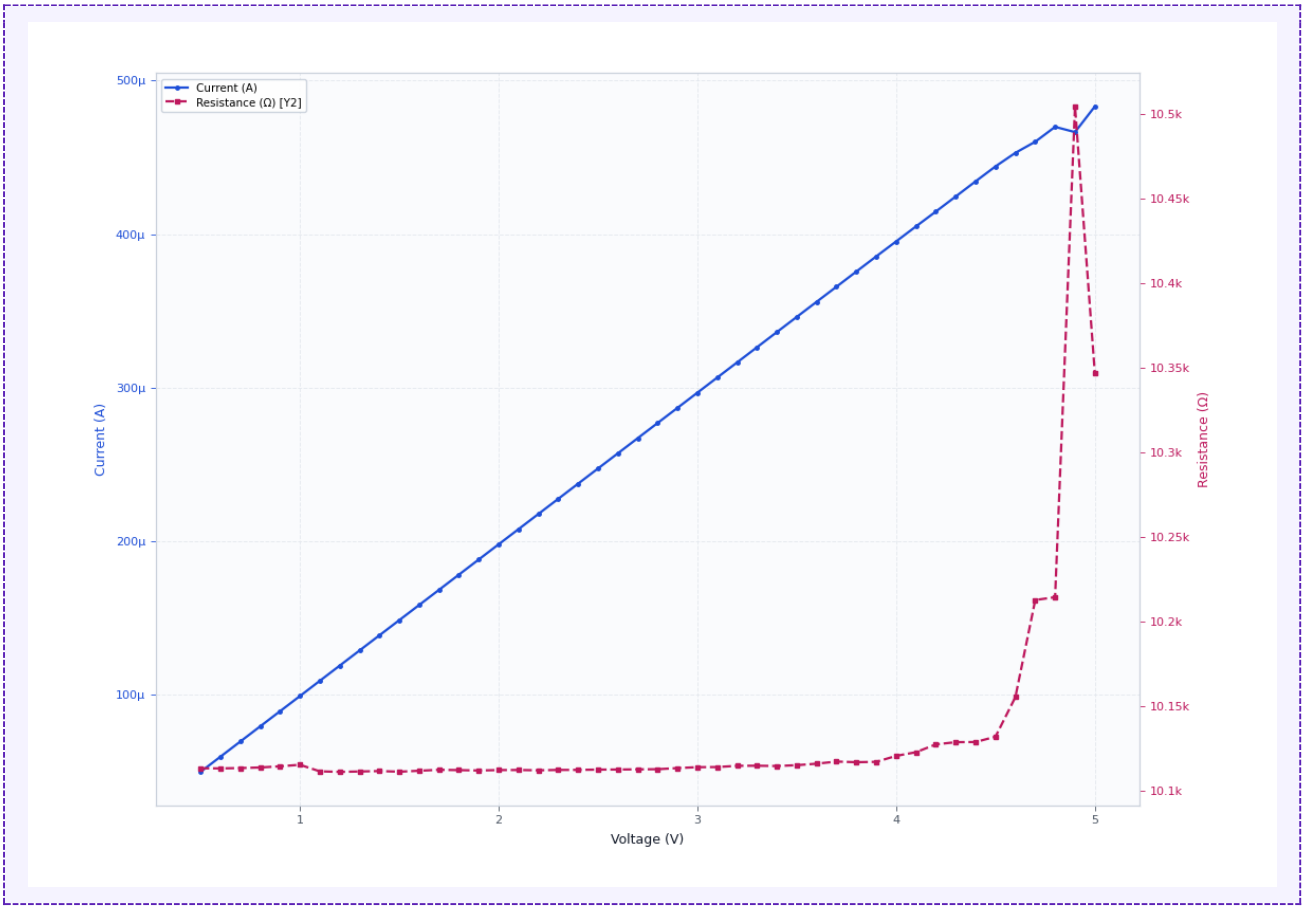

Figure SS-09: Chart Area with Completed I-V Sweep

4.3 Axis Selector Panel

Section	Control	Description
X-Axis	Radio buttons	Select one channel as the horizontal axis. Default: Voltage (V).
Y1-Axis (left)	Checkboxes	One or more channels on the left vertical axis. Default: Current (A).
Y2-Axis (right)	Checkboxes	Secondary right vertical axis — useful for resistance or power alongside current.

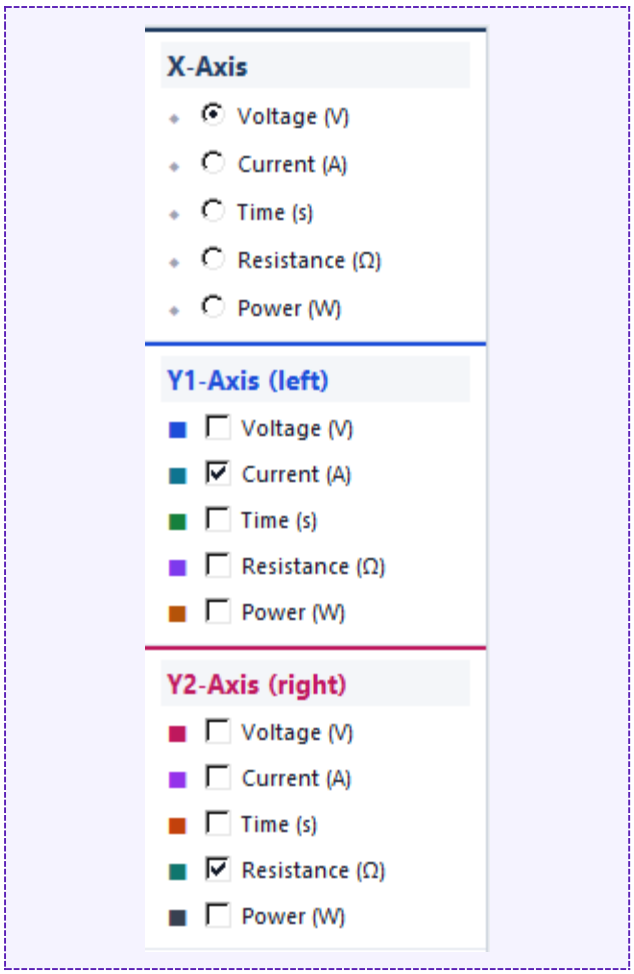

Figure SS-10: Axis Selector Sidebar

5. FET Characterisation Window

The FET Characterisation window is opened from the header bar. It provides a dedicated two-SMU measurement environment for MOSFET characterisation: Gate → SMU1 (Rear), Drain → SMU2 (Rear), Source → GNDU (Rear).

⚠ Warning	Both SMUs must be connected via TSP-Link cable (rear panel TSP-LINK port). Set Node IDs before use: Master = 1, Slave = 2 via MENU → System → TSP-Link Node on the front panel of each unit.

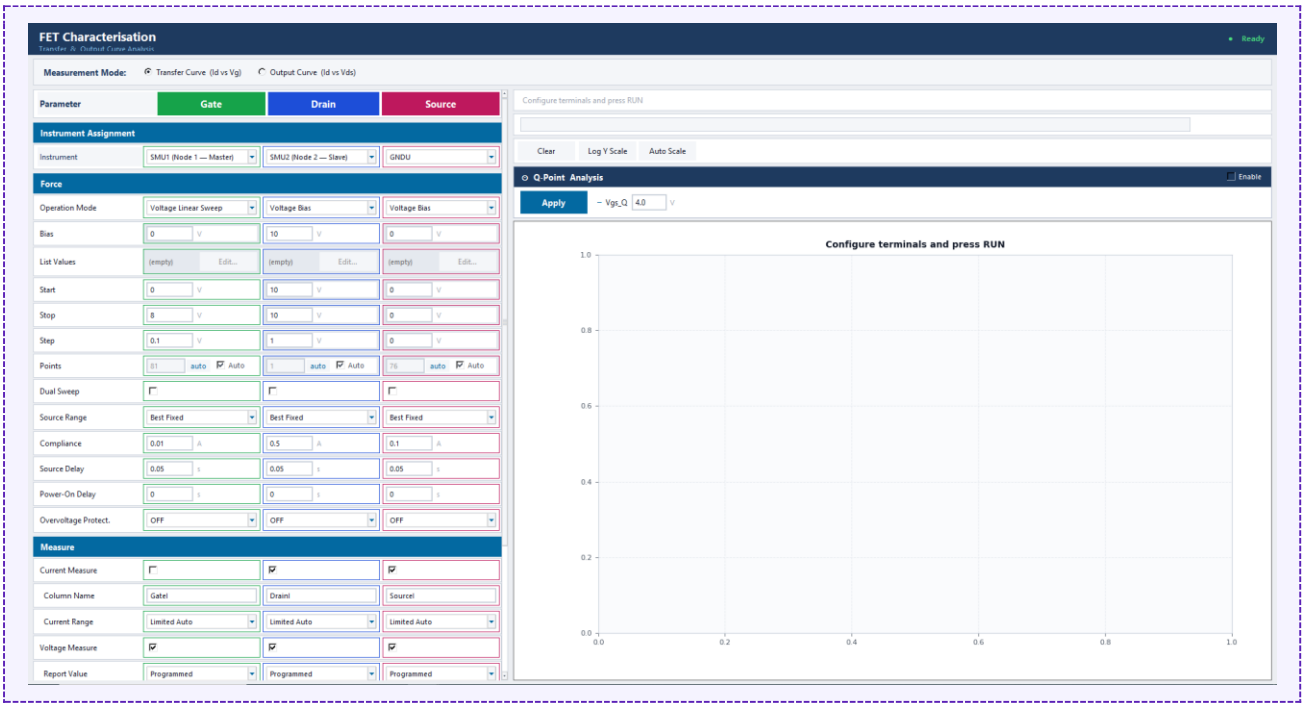

Figure SS-11: Full FET Characterisation Window

5.1 Measurement Mode

Mode	Gate column default	Drain column default	X-axis	Each curve label
Transfer Curve (Id vs Vg)	Voltage Linear Sweep	Voltage Bias	Vgs (V)	Vds = X V
Output Curve (Id vs Vds)	Voltage Bias	Voltage Linear Sweep	Vds (V)	Vg = X V

Switching mode automatically reconfigures the Gate and Drain Operation Mode dropdowns to sensible defaults.

5.2 Parameter Table

Parameter	Gate	Drain	Source
Instrument Assignment			
Instrument	SMU1 (Node 1 — Master)	SMU2 (Node 2 — Slave)	GNDU
Force			
Operation Mode	Voltage Linear Sweep	Voltage Bias	Voltage Bias
Bias	0 V	10 V	0 V
List Values	(empty) Edit...	(empty) Edit...	(empty) Edit...
Start	0 V	10 V	0 V
Stop	8 V	10 V	0 V
Step	0.1 V	1 V	0 V
Points	81 auto ☑ Auto	1 auto ☑ Auto	76 auto ☑ Auto
Dual Sweep	☐	☐	☐
Source Range	Best Fixed	Best Fixed	Best Fixed
Compliance	0.01 A	0.5 A	0.1 A
Source Delay	0.05 s	0.05 s	0.05 s
Power-On Delay	0 s	0 s	0 s
Overvoltage Protect.	OFF	OFF	OFF

Figure SS-12: FET Parameter Table — Force Section

5.2.1 Instrument Assignment

Assign each terminal to NONE, GNDU, SMU1 (Node 1 — Master), or SMU2 (Node 2 — Slave). NONE and GNDU grey out all force/measure parameters for that column.

5.2.2 Operation Mode

Operation Mode	Description
Voltage Bias	Terminal held at a fixed voltage (set in the Bias row)
Voltage Linear Sweep	Terminal sweeps Start → Stop with Step spacing
Voltage List Sweep	Terminal steps through an arbitrary list of voltages
Voltage Log Sweep	Terminal sweeps logarithmically between Start and Stop
Voltage Step	Linear sweep; used for the bias terminal in Output mode
Current Bias / Sweep variants	Same as voltage variants but sources current
Common	Terminal tied to chassis ground through the SMU

5.2.3 Points — Auto and Manual

Each terminal has an Auto/Manual toggle in the Points row:

- **Auto (ticked, default):** Points = |Stop − Start| / |Step| + 1, updated live. Entry field is greyed; blue 'auto' badge shown.
- **Manual (unticked):** Entry field is editable. Sweep uses numpy.linspace(Start, Stop, N) — Step is ignored for point count.

Figure SS-13: Points Row Auto vs Manual

5.2.4 Compliance, Source Delay, and OVP

Terminal	Recommended Compliance	Rationale
Gate	10 mA (0.01 A)	Gate oxide protection — prevents breakdown during transient ringing
Drain	100 mA – 1 A	Set slightly above the maximum expected Id for the gate range used
Source	100 mA	Source is GNDU; this rarely matters unless measuring SourceI

✓ Tip	Source Delay for power MOSFETs (e.g., IRFP240: Ciss ≈ 1400 pF): use 0.10–0.15 s to allow full gate-charge settling. For small-signal MOSFETs 20–50 ms is sufficient.

5.3 Measure Settings and NPLC

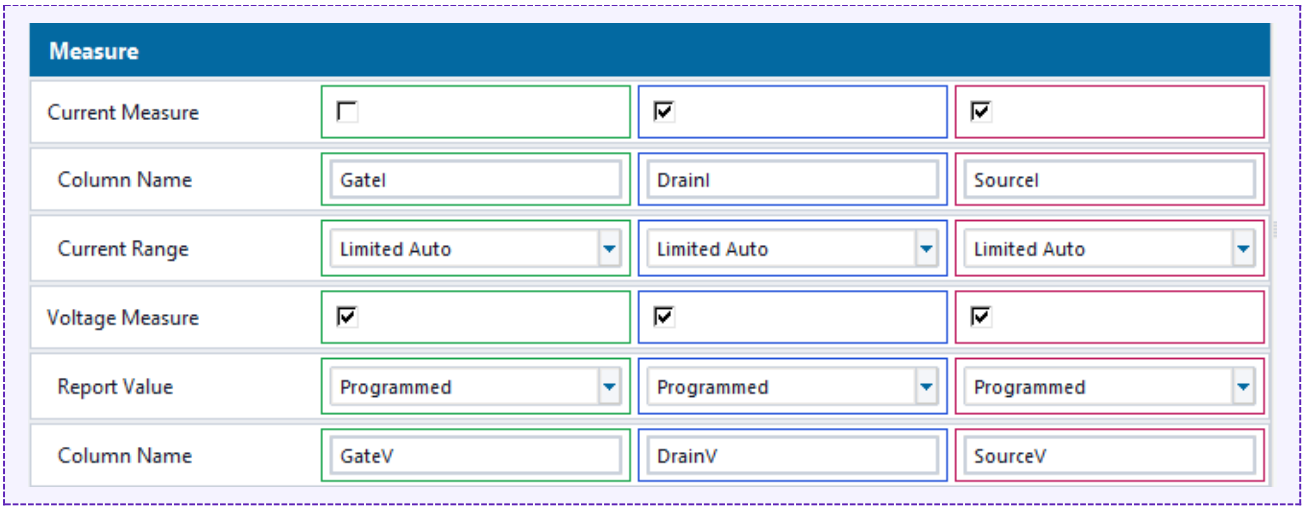

Figure SS-14: FET Measure Section

5.4 Global Settings and Running

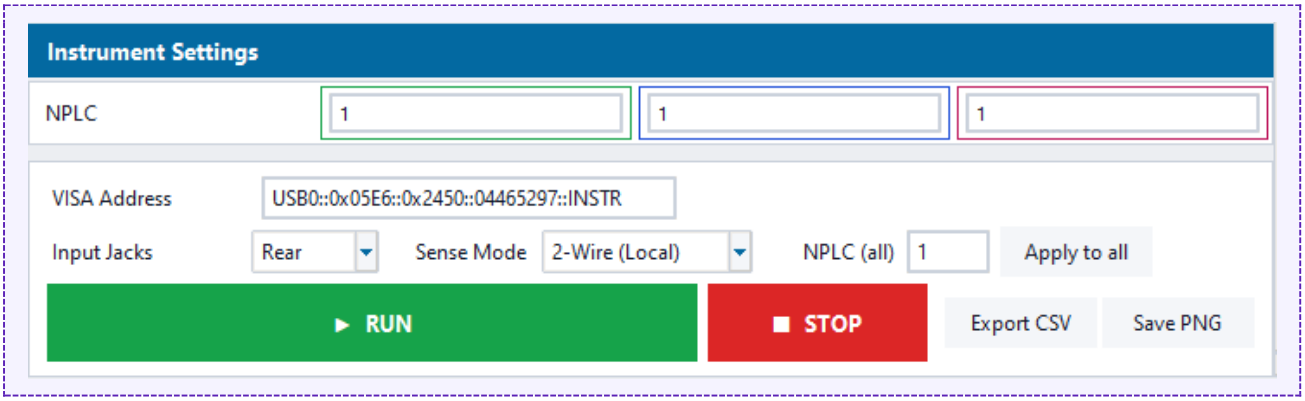

Figure SS-15: FET Bottom Controls Panel

Step-by-step: Running a Transfer Curve

5. Connect: Gate → SMU1 Rear terminals, Drain → SMU2 Rear terminals, Source → GNDU Rear post.
6. Set Node IDs on each instrument front panel: Master = 1, Slave = 2.
7. Enter the Master VISA address.
8. Select Transfer Curve (Id vs Vg) mode.
9. Gate: Start=0, Stop=8 V, Step=0.1 V, Compliance=0.01 A, Delay=0.10 s.
10. Drain: Op Mode = Voltage Bias, Bias=1.5 V, Compliance=0.5 A, Delay=0.10 s.
11. Press RUN. Status shows 'Acquiring…' and the curve builds live.
12. On completion, status shows curve count and auto-saves.

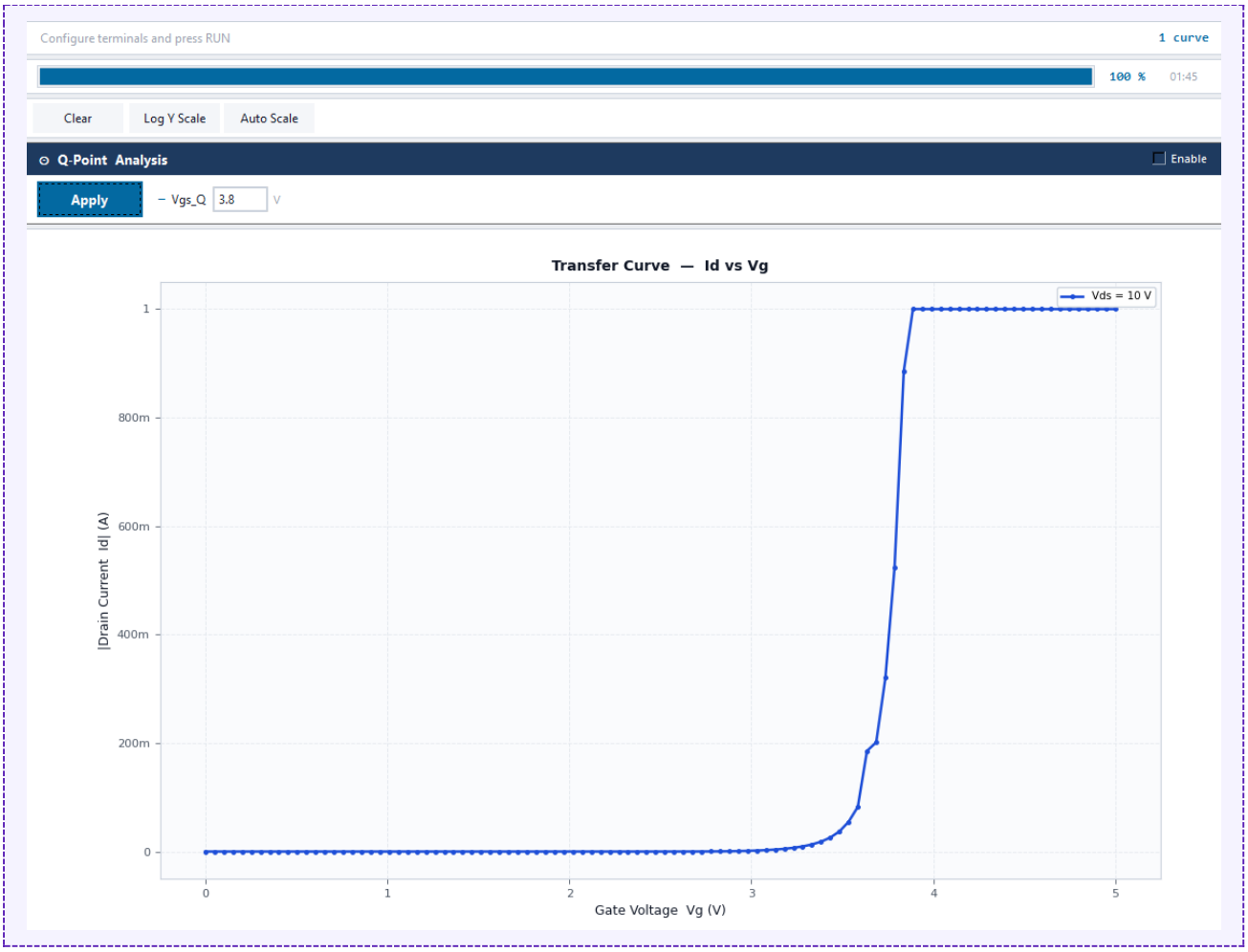

Figure SS-16: Transfer Curve — Measurement Complete

6. Q-Point Analysis

The Q-Point panel sits between the toolbar buttons and the chart in the FET window. Tick the **Enable** checkbox to activate the overlay without requiring a new measurement.

6.1 Transfer Curve Q-Point

Marks a specific gate voltage operating point and computes gm (transconductance) at that point.

Input	Description
Vgs_Q	The gate-to-source quiescent voltage at which to evaluate the Q-point (Volts)

Output on chart:

- Gold star marker (★) at (Vgs_Q, Id_Q)
- Dashed yellow crosshair lines from both axes to the marker
- **Annotation box showing:** Vgs, Id, and gm (Siemens)
- gm computed as dId/dVgs using numpy.gradient on the measured curve

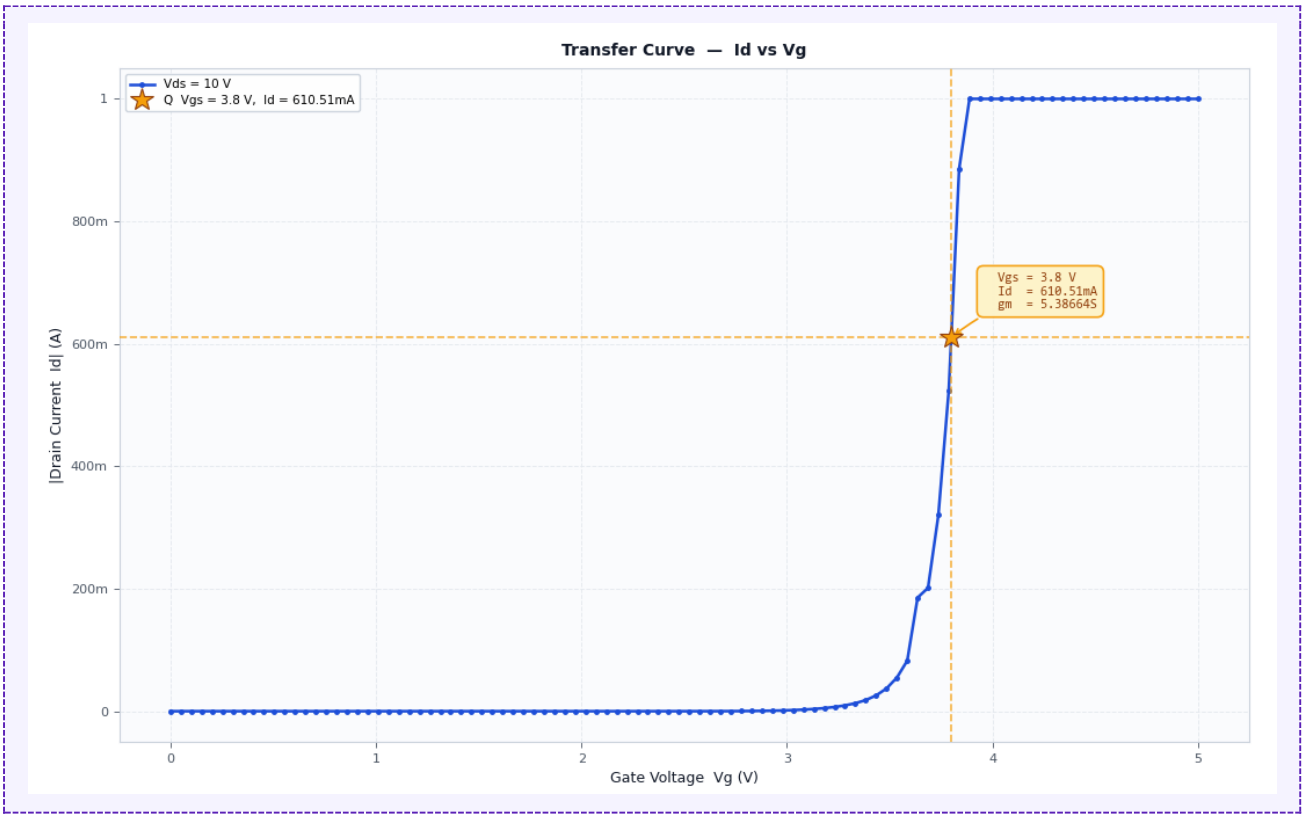

Figure SS-17: Transfer Curve with Q-Point Overlay

6.2 Output Curve Q-Point

Draws a DC load line, finds its intersection with a selected Vg curve, and plots the constant dissipation hyperbola through the operating point.

Input	Description
Vdd	Supply voltage — x-axis intercept of the load line (Id = 0 at Vds = Vdd)
RL	Load resistance (Ω) — sets the slope. Id = Vdd/RL at Vds = 0.
Vg curve	The Vgs bias curve to intersect — auto-populated from measured data

Output on chart:

- Dash-dot gold load line from (Vdd, 0) to (0, Vdd/RL) with intercept labels
- Gold star marker (★) at the intersection point (Vds_Q, Id_Q)
- **Red dotted Pd hyperbola:** Id = Pd_Q / Vds through the Q-point (equal-dissipation locus)
- **Annotation box:** Vg curve label, Vds_Q, Id_Q, Pd_Q

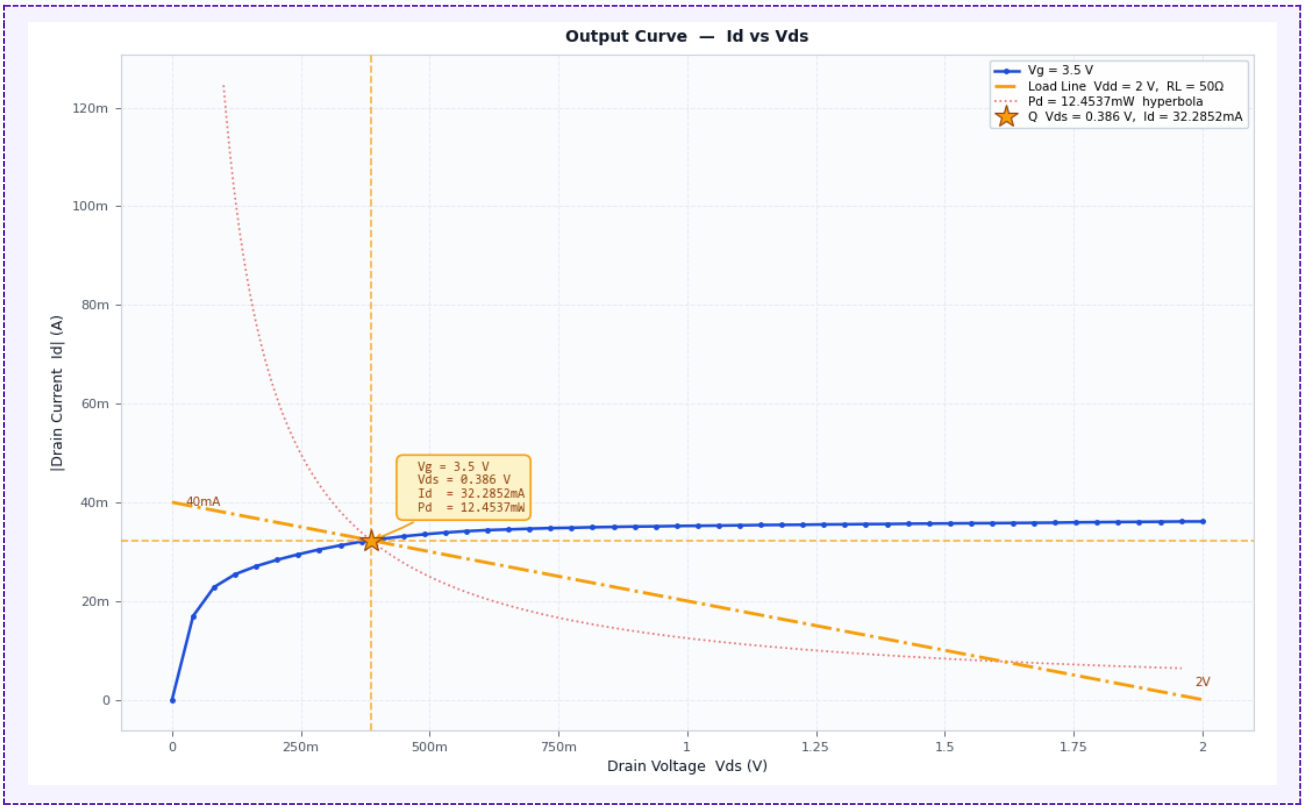

Figure SS-18: Output Curve with Load Line and Q-Point

⚠ Warning	The Q-point is found at the intersection of the load line with the selected Vg curve. If no intersection is found (e.g., Vdd is smaller than the sweep range), 'No Q-point intersection' appears in the result bar.

7. Data Export

7.1 CSV Format — Main Window

```
# Keithley 2450  I-V Studio  v3.2
# Sample ID, IRFP240
# Date/Time, 01/01/2025  14:32:17
# Mode, Voltage Sweep
# NPLC, 1   # Compliance, 0.01   # Terminals, Rear   # Sense, 2-Wire

Time (s), Voltage (V), Current (A), Resistance (Ohm), Power (W)
0.000000, 0.000000000, 0.000012450, 0.000000, 0.000000000
```

# Keithley 2450 I-V Studio v3.0			
# Sample ID	DUT_001		
# Date/Time	26-02-2026 12:44		
# Mode	Voltage Sweep		
# NPLC	1		
# Compliance	0.01		
# Terminals	Front		
# Sense	2-Wire		
Time (s)	Voltage (V)	Current (A)	Resistance (Ω)
0	0.49981308	4.94E-05	10113.5
0.14211	0.599773407	5.93E-05	10113.4
0.284172	0.699619293	6.92E-05	10113.6
0.426239	0.799823761	7.91E-05	10114
0.5683	0.899612427	8.89E-05	10114.7
0.710365	0.999782562	9.88E-05	10115.6
1.076689	1.09984207	0.000108769	10111.7
1.21877	1.1997757	0.000118656	10111.4
1.360836	1.29997635	0.000128563	10111.6
1.502916	1.39987946	0.000138439	10111.9
1.645003	1.50000381	0.000148347	10111.5
1.787057	1.5998764	0.000158215	10112
1.929108	1.69976425	0.000168084	10112.6

Figure SS-19: CSV Export — Main Window File Opened in Excel

7.2 CSV Format — FET Window

```
# Keithley 2450  I-V Studio  v3.2  –  FET Characterisation
# Mode, Transfer Curve  (Id vs Vg)
```

```
# Gate, SMU1 (Node 1 – Master), Voltage Linear Sweep
# Date, 01/01/2025  14:32:17

Vg (V), Id  Vds = 1.5 V (A), Id  Vds = 3.5 V (A)
0.000000, 0.000000012, 0.000000015
```

# Keithley 2450 I-V Studio v3.0 — FET Characterisation								
# Mode	Output Curve (Id vs Vds)							
# Gate	SMU1 (Node 1 —	Voltage Linear Sweep						
# Drain	SMU2 (Node 2 —	Voltage Linear Sweep						
# Source	GNDU	Voltage Bias						
# Date	26-02-2026 14:43							
Vds (V)	Id Vg = 3 V (A)	Id Vg = 3.2 V (A)	Id Vg = 3.4 V (A)	Id Vg = 3.6 V (A)	Id Vg = 3.8 V (A)	Id Vg = 4 V (A)	Id Vg = 4.2 V (A)	Id Vg = 4.4 V (A)
0	-2.80E-07	-2.01E-08	-2.22E-08	-1.63E-07	-2.01E-07	-2.18E-06	-1.93E-05	-1.93E-05
0.0408163	0.000774904	0.002130371	0.005982447	0.006438251	0.029313907	0.0862673	0.150119781	0.15230751
0.0816327	0.001108361	0.003238369	0.010785103	0.051350966	0.153774261	0.259254456	0.300197601	0.304674149
0.122449	0.001266889	0.003964836	0.012346208	0.057206675	0.259969711	0.422702789	0.450183868	0.457054138
0.163265	0.00132392	0.004469394	0.013474777	0.060210839	0.298629761	0.562294006	0.60001564	0.609388351
0.204082	0.00134371	0.004793778	0.014408156	0.062419578	0.3189888	0.698495865	0.749666214	0.761659622
0.244898	0.001351684	0.00498247	0.015194669	0.064096622	0.330863953	0.83205986	0.899148941	0.913883209
0.285714	0.001357486	0.0050865	0.015857264	0.065598436	0.33905983	0.962024689	1.00004387	1.00004578
0.326531	0.001357643	0.005137743	0.016346857	0.066937394	0.345434189	1.00003433	1.00003433	1.00003815
0.367347	0.001358278	0.005166104	0.016745374	0.068107598	0.351224899	1.0000267	1.00003052	1.00002861
0.408163	0.001359306	0.005182554	0.017041862	0.069104284	0.355516434	1.0000267	1.0000248	1.0000267
0.44898	0.001359964	0.005193761	0.017266139	0.070076905	0.35959053	1.00002098	1.00002289	1.00002289
0.489796	0.001360441	0.005201031	0.017426044	0.070902683	0.363294601	1.00002098	1.00001907	1.00002289
0.530612	0.001361048	0.005209302	0.017521799	0.071605459	0.366502762	1.00001907	1.00001526	1.00001717
0.571429	0.00136123	0.005213147	0.017596871	0.072167419	0.369754791	1.00001526	1.00001526	1.00001335
0.612245	0.001361903	0.005216093	0.017664939	0.072708897	0.372585297	1.00000954	1.00000954	1.00000572
0.653061	0.001362083	0.005222325	0.017710537	0.073186383	0.375156403	1.00000954	1.00000763	1.00000954
0.693878	0.001362894	0.005218643	0.017740369	0.07360521	0.377723694	1.00000381	1.00000381	1.00000572
0.734694	0.001362929	0.005221937	0.017772213	0.07396175	0.380069733	0.999998093	1	1
0.77551	0.001362111	0.005230367	0.017793521	0.074363612	0.382415771	1.00000191	1	1.00000191
0.816327	0.00136303	0.0052254	0.017819449	0.074565589	0.384552002	1	1	0.999998093
0.857143	0.001363603	0.005229364	0.017855346	0.074790224	0.386655808	0.999990463	0.999994278	0.999992371

Figure SS-20: FET CSV Export Opened in Excel

8. Typical Measurement Workflows

8.1 Diode I-V Curve

13. Connect diode: anode → FORCE HI, cathode → FORCE LO (Rear terminals).
14. Set Source Mode: Voltage Sweep, Start=0, Stop=1.0 V, Step=0.01 V.
15. Set Compliance=0.05 A, Source Delay=0.05 s, Input Jacks: Rear.
16. Confirm VISA address. Press RUN.

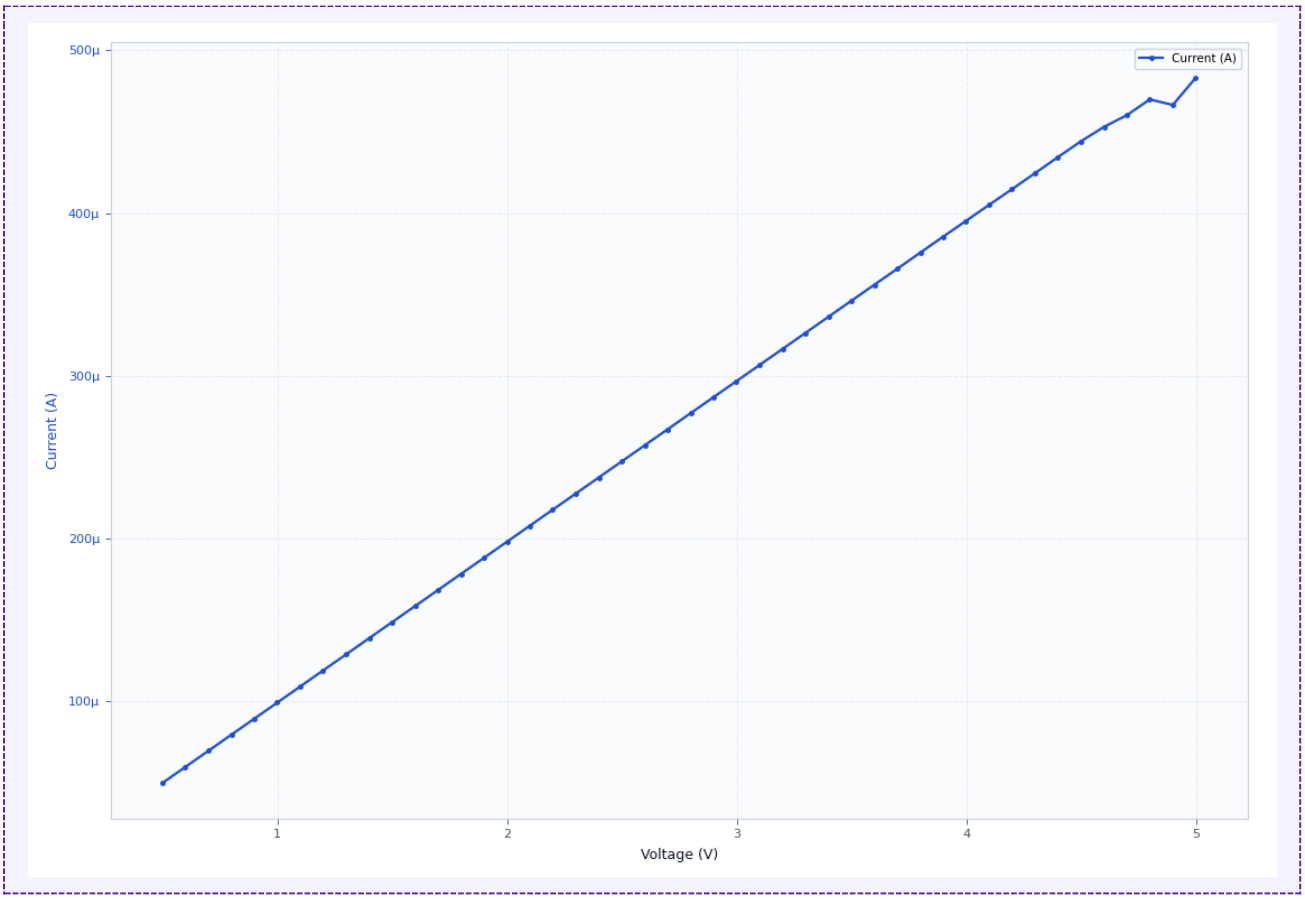

Figure SS-21: Completed Diode I-V Curve

8.2 MOSFET Transfer Curve

17. Open FET window. Select Transfer Curve.
18. Gate: Start=0, Stop=8 V, Step=0.1 V, Compliance=0.01 A, Delay=0.10 s.
19. Drain: Voltage Bias, Bias=1.5 V, Compliance=0.5 A.
20. Press RUN. One curve is produced at Vds=1.5 V.

✓ **Tip**	For a family of Transfer Curves at multiple Vds, set Drain Op Mode to Voltage Step: configure Drain Start/Stop/Step to the desired range. Each step produces one curve.

8.3 MOSFET Output Curve

21. Open FET window. Select Output Curve.
22. Drain: Start=0, Stop=15 V, Step=0.2 V, Compliance=0.5 A.
23. Gate: Voltage Step, Start=2 V, Stop=6 V, Step=1 V.
24. This produces 5 curves: Vgs = 2, 3, 4, 5, 6 V.
25. Press RUN.

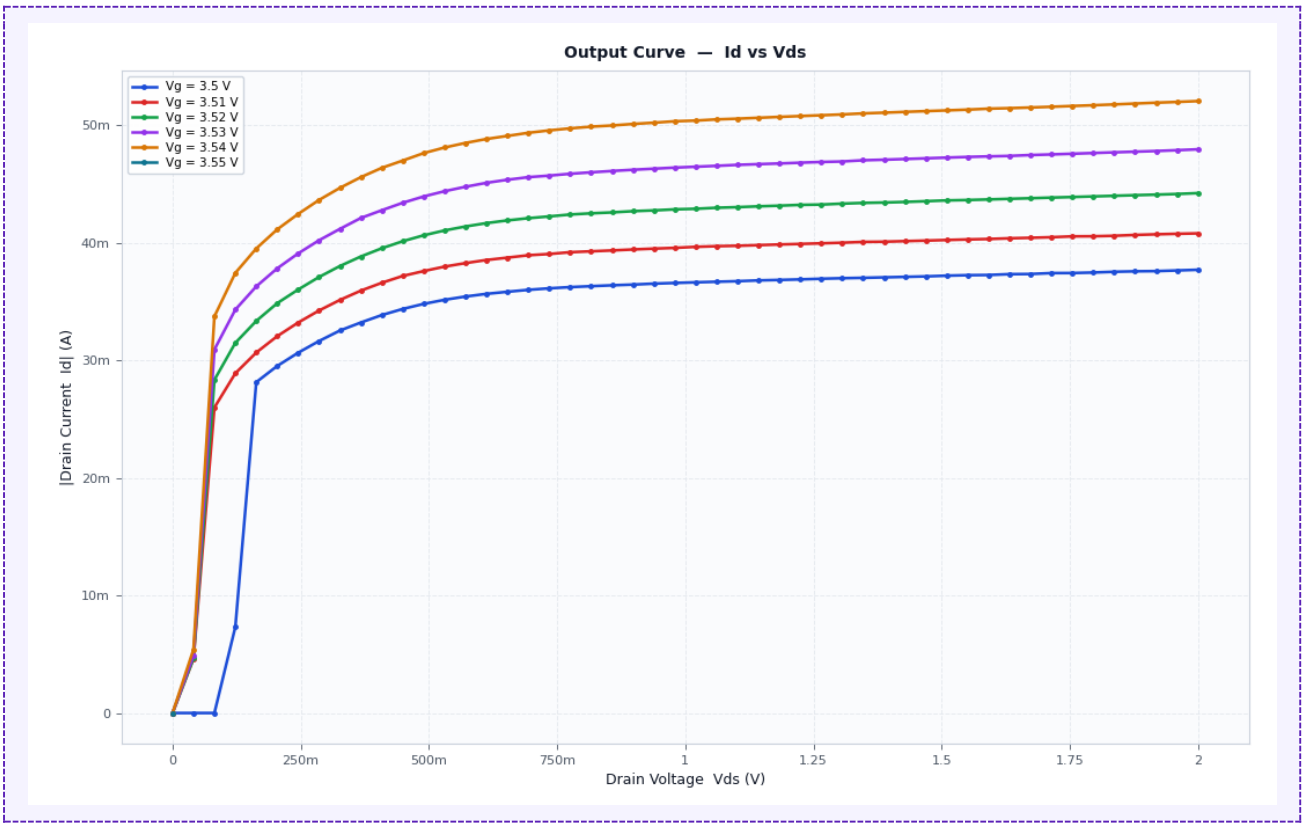

Figure SS-22: Completed Output Curve — Family of Curves

8.4 Q-Point Analysis on Output Curves

26. Complete the Output Curve measurement (Section 8.3).
27. In the Q-Point panel, tick Enable.
28. Set Vdd=20 V, RL=100 Ω.
29. Select the Vg=4 V curve from the dropdown.
30. Press Apply. Load line and Q-point appear on the chart.
31. Read Vds_Q, Id_Q, Pd_Q from the result bar and annotation box.
32. Press Save PNG to export the annotated chart for your report.

8.5 Hysteresis Measurement

33. Enable Dual Sweep in the FET window Dual Sweep row (or main window Dual Sweep checkbox).
34. Configure the sweep normally. Press RUN.
35. The dataset contains both the forward and reverse passes as one curve.

i Note	For TFTs and organic semiconductors, the Transfer Curve in log-Y scale is particularly diagnostic for hysteresis from trap states. Use the Log Y Scale button in the FET chart toolbar.

9. Troubleshooting

Symptom	Likely Cause	Resolution
VISA error / instrument not found	Wrong VISA address or VISA runtime not installed	Open NI-MAX or Keysight Connection Expert, verify the resource string.
TSP-Link not online error	TSP-Link cable missing or node IDs wrong	Check cable. Verify Node ID=1 on Master and Node ID=2 on Slave.
All current values flat at compliance	Device hitting compliance	Increase Compliance. Raise Source Delay to 0.2 s.
Noisy, non-monotonic Transfer Curve	Gate charge not settling before measurement	Increase Gate Source Delay to 0.15–0.20 s.
Blank plot after sweep	Axis limits do not include the data range	Click Auto Scale. Check Y-axis selector — ensure Current (A) is ticked on Y1.
Q-point shows 'No intersection'	Load line does not cross the selected Vg curve	Increase Vdd or reduce RL so the load line crosses the saturation region.
Auto-save failed warning	Save directory missing or read-only	Create D:\KEITHLEY 2450\Data or use Export CSV manually.
Sweep timeout error	Too many points or NPLC too high	Reduce Points, reduce NPLC, or reduce Source Delay.

10. Quick Parameter Reference

10.1 Main Window Defaults

Parameter	Default	Notes
Source Mode	Voltage Sweep	
Start / Stop / Step	0 / 5 V / 0.1 V	
Points	51	Auto-calculated from step
Compliance	0.01 A	10 mA — increase for higher-current devices
Source Delay	0.20 s	
NPLC	1	20 ms per point at 50 Hz
Auto Zero	Once	
Input Jacks	Rear	
Sense Mode	2-Wire	
Save Directory	D:\KEITHLEY 2450\Data	Create if it does not exist

Appendix A — Glossary

Term	Definition
Compliance	Maximum current (for V source) or voltage (for I source) the SMU will allow — acts as a protective clamp.
GNDU	Ground Unit — the chassis ground reference terminal of the Keithley 2450.
gm	Transconductance. Small-signal ratio dId/dVgs (Siemens). Indicates the gain available from the transistor.
Id	Drain current — the primary measured quantity in FET characterisation.
NPLC	Number of Power Line Cycles. Sets integration time per measurement. 1 NPLC = 20 ms at 50 Hz.
Q-point	Quiescent operating point. The steady-state DC bias conditions around which a circuit operates.
SMU	Source Measure Unit — simultaneously sources voltage or current and measures the complementary quantity.
Pd	Power dissipation at the Q-point: Pd = Vds × Id. Watts.
TSP	Test Script Processor — scripting language built into Keithley 2450 instruments.
TSP-Link	Keithley proprietary RJ-45 instrument bus allowing multiple SMUs to be synchronised from a single VISA connection.
VISA	Virtual Instrument Software Architecture — standard interface layer over USB, GPIB, LAN, etc.
Vgs / Vth	Gate-to-source voltage / threshold voltage. Gate voltage above which the MOSFET begins to conduct.

— End of Manual —